

Giordano Alvari

Senior AI / ML Engineer

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Senior AI / ML Engineer with 6+ years delivering production machine-learning and LLM systems. Experience spans architecture, backend implementation, evaluation, deployment, small-team coordination and cross-country delivery. Background in computational biology and statistical modelling, with a peer-reviewed publication in *Genome Biology*.

EXPERIENCE

Enel — Senior AI / ML Engineer Florence, Italy · Jun 2023–Present

- Design and deliver tool-using LLM and retrieval systems integrated with operational monitoring and ticketing workflows.
- Lead architecture, backend implementation, evaluation and deployment across Python services, AWS Bedrock, FastAPI and Kubernetes.
- Develop hierarchical customer-base forecasts with conformal intervals and scenario simulation for planning workflows.
- Coordinate technical delivery for small teams and collaborate with operational and executive stakeholders across Italy and Spain.

Enel — Machine Learning Engineer Rome, Italy · Sep 2020–Jun 2023

- Built predictive systems on distributed customer data, owning ETL, feature engineering, training and serving.
- Delivered scenario forecasting, customer segmentation, cross-selling and battery anomaly-detection models.
- Contributed to MLOps standards using MLflow and Airflow.

DKFZ — Computational Biologist Heidelberg, Germany · Sep 2019–Sep 2020

- Designed a reproducible single-cell eQTL mapping pipeline in collaboration with EMBL; work published in *Genome Biology*.

SELECTED PROJECTS

GridCast — Leakage-safe probabilistic energy forecasting with public PJM load, ERA5 weather, LightGBM and conformal calibration.

RadixForge — C++ radix-tree KV-cache orchestrator for reusable prompt prefixes through native llama.cpp APIs.

LocalLLM Bench — Reproducible local-inference evaluation across quality, throughput and memory with seeded load and run-level intervals.

Agent Memory Layer — OpenAI-compatible C++17 proxy for persistent local agent memory with llama.cpp embeddings and SQLite retrieval.

SKILLS

LLM systems: agents, RAG, tool/function calling, Text2SQL, retrieval evaluation, LangChain, LlamaIndex, AWS Bedrock, pgvector, llama.cpp.

Machine learning: forecasting, conformal prediction, PyTorch, scikit-learn, XGBoost, PySpark, Nixtla, skforecast.

Engineering: Python, C++, SQL, FastAPI, Pydantic, Docker, Kubernetes, AWS, MLflow, Airflow, CI/CD, MongoDB, SQLite.

Delivery: technical architecture, end-to-end implementation, model evaluation, team coordination, AI governance and stakeholder communication.

EDUCATION, PUBLICATION & LANGUAGES

MSc, Quantitative and Computational Biology — University of Trento 2017–2019

BSc, Biomolecular Sciences and Technology — University of Trento 2014–2017

Publication: Cuomo, A. S. E., Alvari, G., et al. “Optimizing expression quantitative trait locus mapping workflows for single-cell studies.” *Genome Biology*, 2021. doi.org/10.1186/s13059-021-02407-x.

Languages: Italian — native · English — C1 · Spanish — B1.

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